Introduction to Probability and Applied Statistics

Week 8: Linear Regression I

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Roadmap

Relating two variables

Where we are

The difference in means is a regression

From conditional means to a line

Least squares

Interpreting the line

How well does the line fit?

Wrapping up

Why relate two variables?

- Do cities with more renters have lower council-election turnout?
- Do women in a legislature speak for less time in committee hearings than men do?
- Do older citizens vote at higher rates than younger ones?
- Do women rate Kamala Harris more warmly than men do?
- How do ratings of Harris change from strong Democrats to strong Republicans?

Each asks: how does the average of Y differ across values of X?

Outcome and explanatory variables

X

The variable we use to describe or predict *Y*.

explanatory variable
independent variable
covariate
regressor

Y

The variable we want to describe or predict.

outcome
dependent variable
response variable
regressand

Where we are going

The rest of the course is about one tool: **linear regression**.

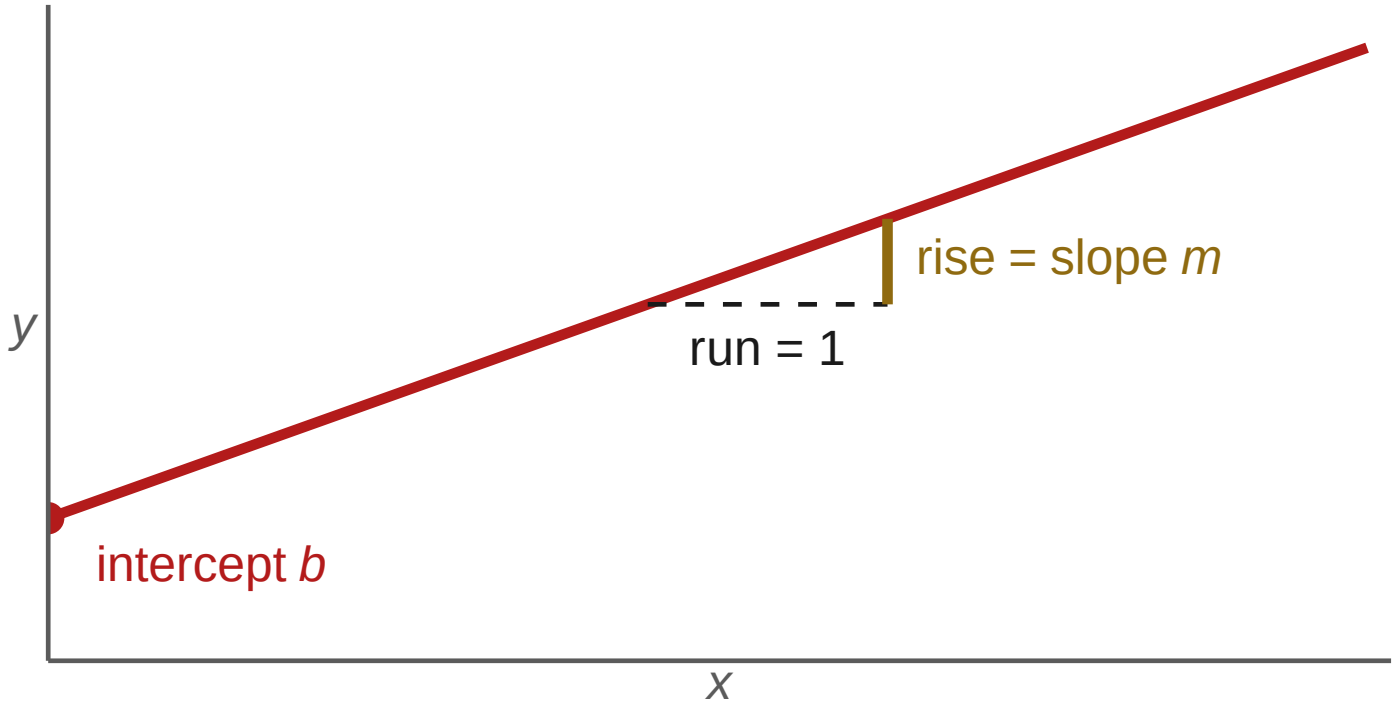
- Fitting a line to two variables, interpreting it, measuring its fit
- How the fitted line varies from sample to sample
- More than one X; group indicators, interactions, curves
- Tests and confidence intervals for regression
- Presenting results; regression and causal questions

The best guess about *Y*

For any value of ***X***, what is the best guess about ***Y***?

The simplest way to relate two variables is a line:

$$y = mx + b$$



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The mean is the best constant prediction

THE MEAN AS A PREDICTION

Predict the same number c for every unit. The prediction with the smallest total squared error is the sample mean:

$$\bar{Y} = \arg\min_{c} \sum_{i} (Y_i - c)^2, \text{ uniquely.}$$

The conditional mean

CONDITIONAL MEAN

The mean of Y among the units with $X = x$:

$$\bar{Y}_{X=x} = \frac{\sum_i Y_i \mathbf{1}\{X_i = x\}}{\sum_i \mathbf{1}\{X_i = x\}}.$$

Read as a function of x, $\bar{Y}_{X=x}$ is the **conditional mean function**.

The difference between two groups

DIFFERENCE IN MEANS

For two groups A and B, the difference in average Y between A and B is

$$\bar{Y}_A - \bar{Y}_B.$$

Covariance and correlation

SAMPLE COVARIANCE AND CORRELATION

$$\widehat{\mathrm{Cov}}(X, Y) = \frac{1}{n-1} \sum_i (X_i - \bar{X})(Y_i - \bar{Y}), \qquad \hat{r} = \frac{\widehat{\mathrm{Cov}}(X, Y)}{s_X \, s_Y}.$$

s_X and s_Y are the sample standard deviations, and $s_X^2 = \widehat{\mathrm{Cov}}(X, X)$.

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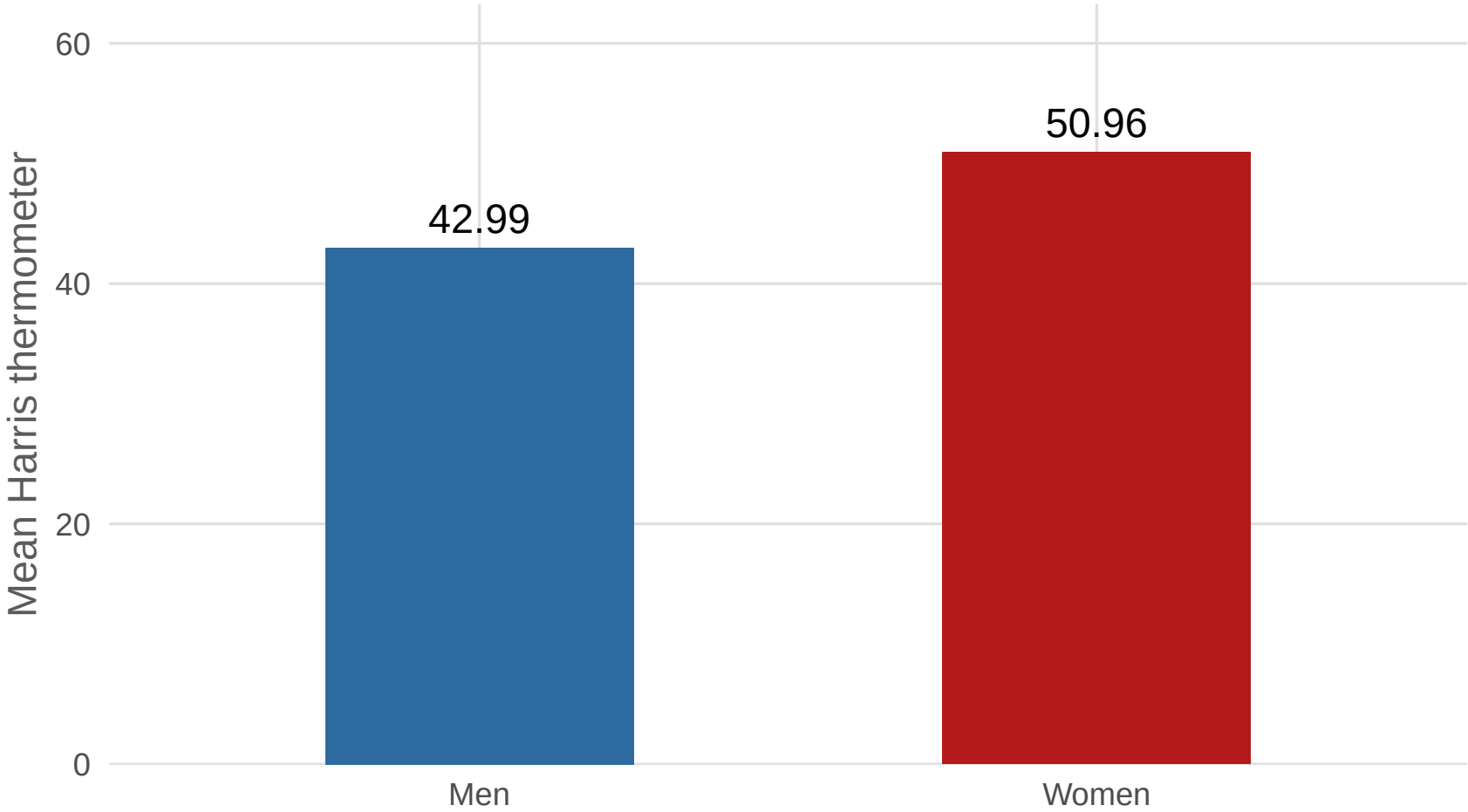
Wrapping up

Two survey items

Variable	ANES 2024 item	Values
`harris`	Feeling thermometer: Kamala Harris (V241156)	0–100
`female`	"What is your sex?" (V241550)	1 = female, 0 = male

Pre-election wave, unweighted, $n = 5{,}455$ respondents who answered both.

Two group means



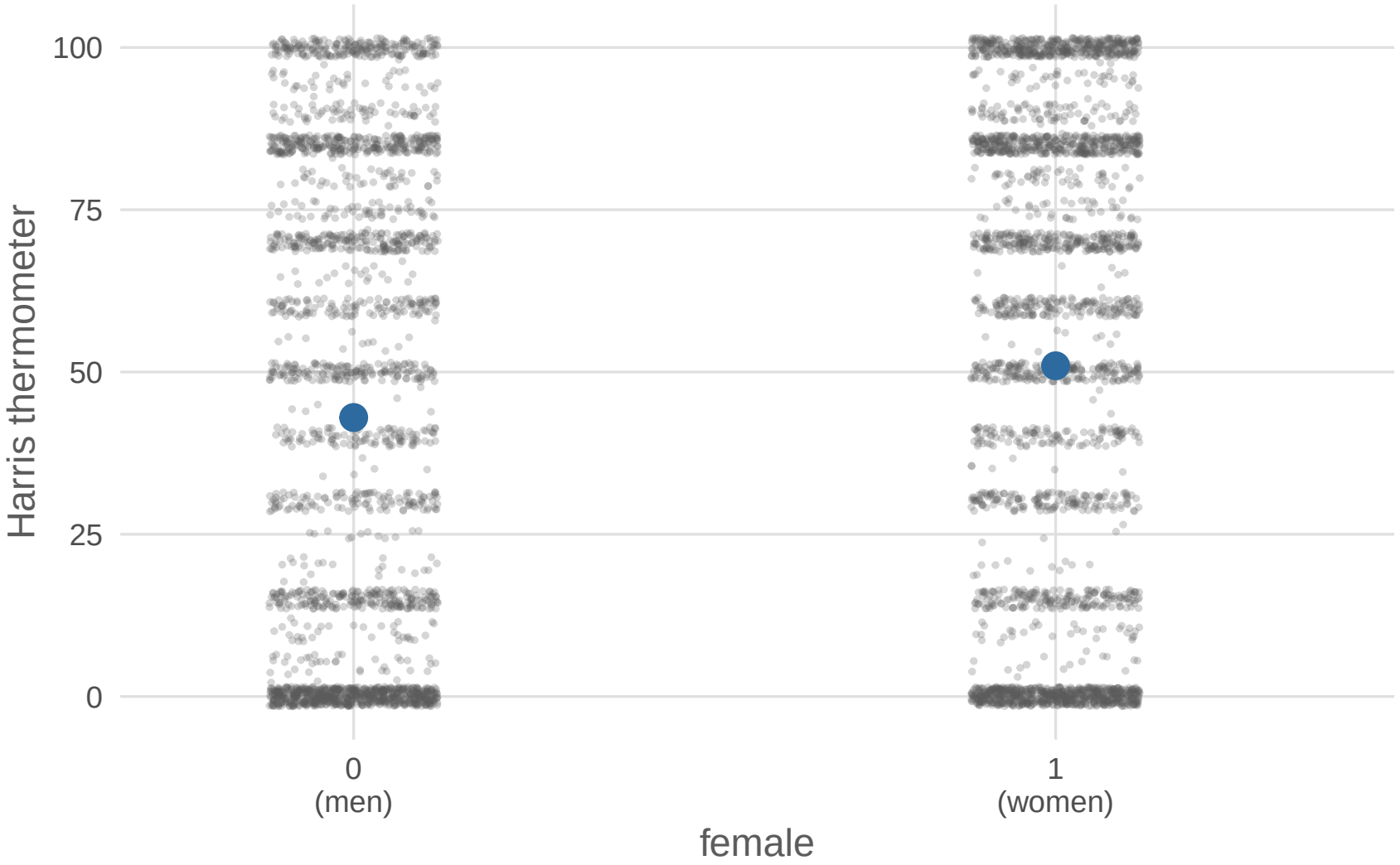
$$\bar{Y}_{\text{women}} - \bar{Y}_{\text{men}} = 50.96 - 42.99 = 7.97$$

Sex as a 0/1 variable

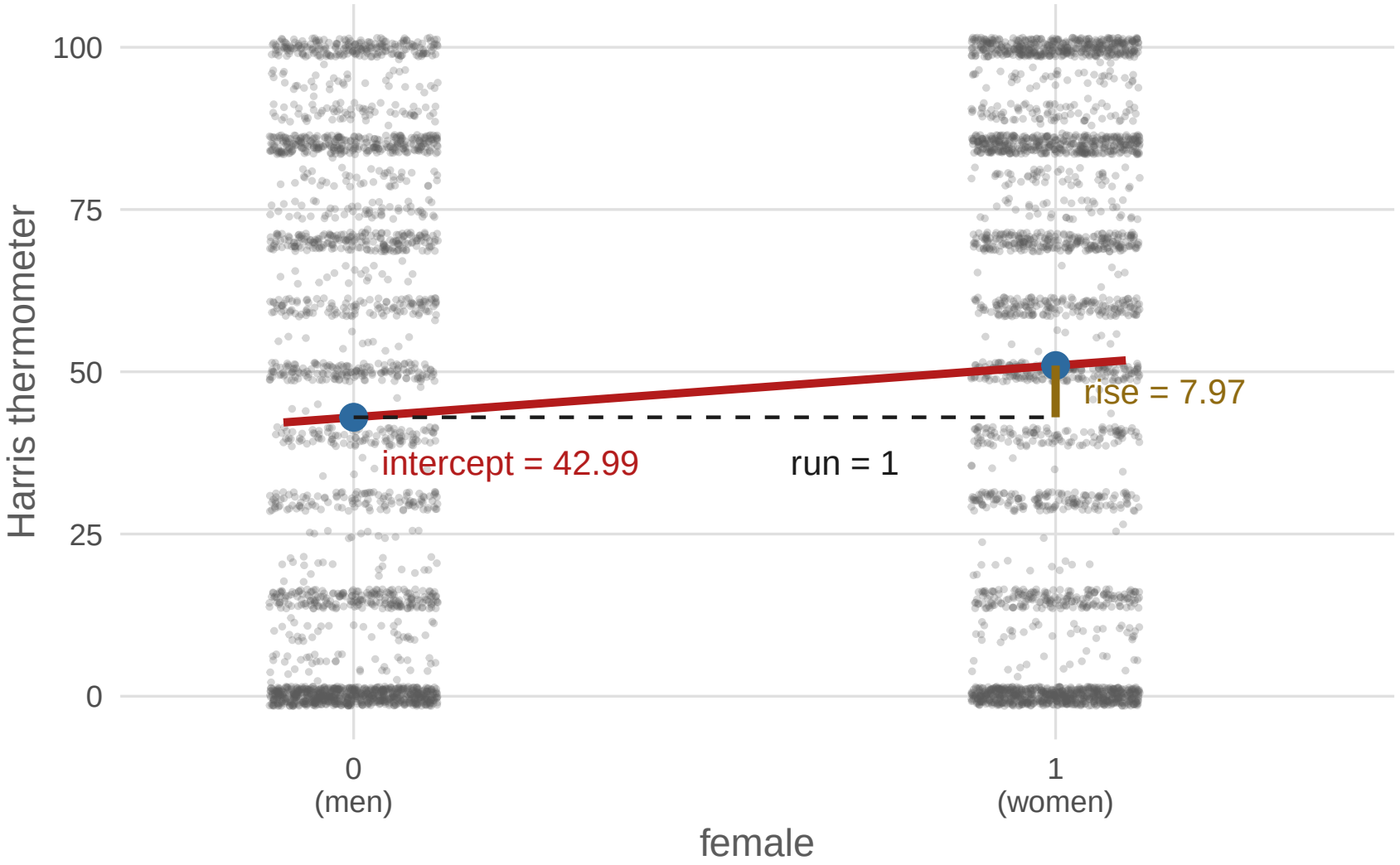
id	harris	female
140001	0	0
140002	50	1
140003	90	1
140004	50	0
140005	5	0
140006	85	1

A variable coded 1 for one group and 0 for the other is a **binary** or **dummy** variable.

The ratings by group



A line through the two means



Asking R for the line

```
fit <- lm(harris ~ female, data = anes)
coef(fit)
```

```
(Intercept)      female
  42.989345    7.967519
```

By hand and by `lm()`

BY HAND

$\bar{Y}_{X=0} = 42.99$

$\bar{Y}_{X=1} - \bar{Y}_{X=0} = 7.97$

`lm()`

intercept = 42.99

slope = 7.97

Regression on a binary variable

A LINE THROUGH TWO GROUP MEANS

When X takes only the values 0 and 1,

$$\text{intercept} = \bar{Y}_{X=0}, \qquad \text{slope} = \bar{Y}_{X=1} - \bar{Y}_{X=0}.$$

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Party identification

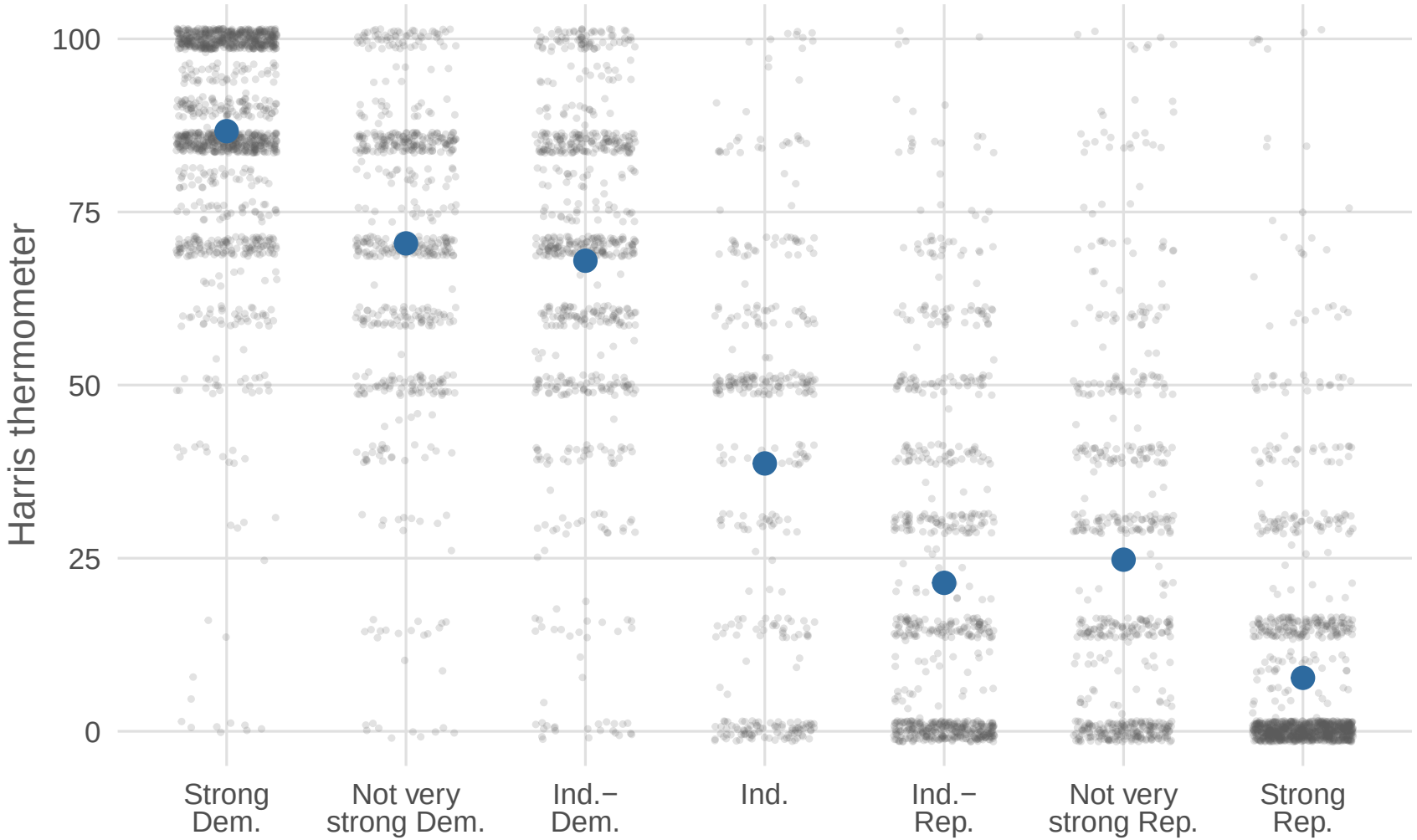
`pid`	Party ID (V241227x)
1	Strong Democrat
2	Not very strong Democrat
3	Independent-Democrat
4	Independent
5	Independent-Republican
6	Not very strong Republican
7	Strong Republican

`harris`	Harris thermometer (V241156)
0–49	Unfavorable, cold
50	Neither warm nor cold
51–100	Favorable, warm

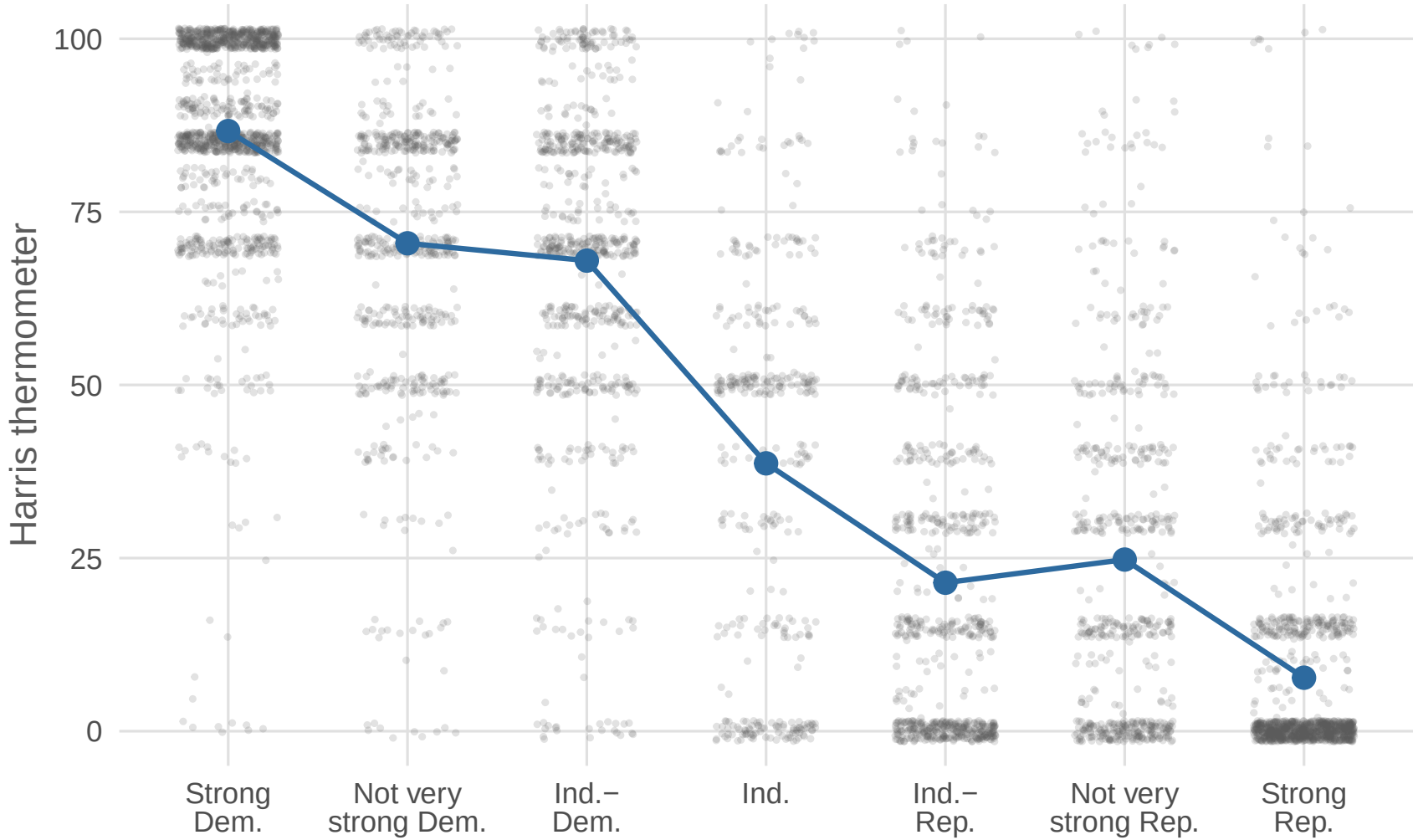
Harris ratings by party ID

x	1	2	3	4	5	6	7
$\bar{Y}_{X=x}$	86.7	70.5	68.0	38.7	21.4	24.8	7.7
n	1,312	613	713	377	714	575	1,163

Every respondent, and the seven means



The conditional mean function



What should f look like?

$$\mathbb{E}[Y \mid X = x] = f(x)$$

$\mathbb{E}[Y \mid X = x]$ is the population quantity that $\bar{Y}_{X=x}$ estimates.

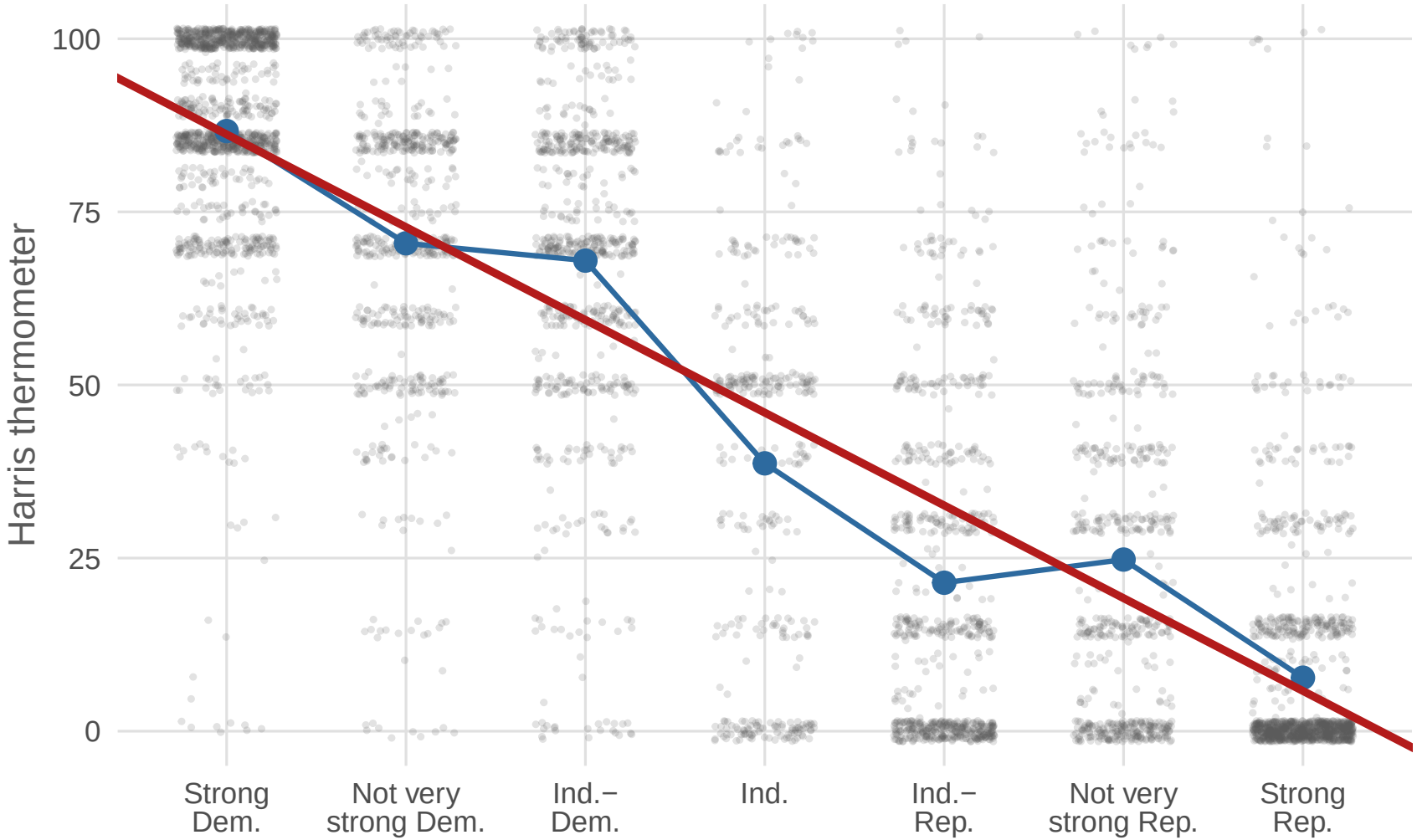
x	1	2	3	4	5	6	7
$f(x)$	$f(1)$	$f(2)$	$f(3)$	$f(4)$	$f(5)$	$f(6)$	$f(7)$
$\bar{Y}_{X=x}$	86.7	70.5	68.0	38.7	21.4	24.8	7.7

A LINEAR CONDITIONAL MEAN

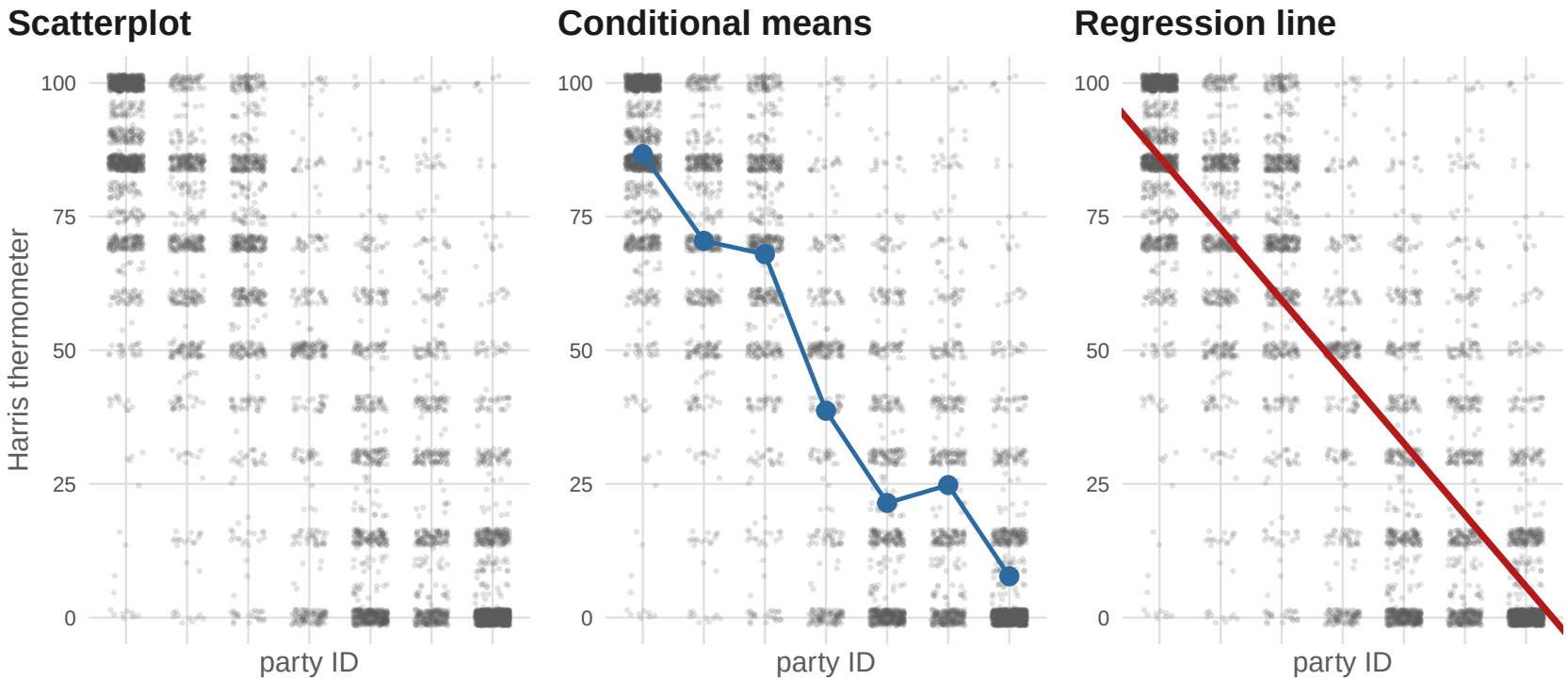
$$f(x) = \beta_0 + \beta_1 x$$

β_0 is the **intercept**, β_1 the **slope**. Together they are the **coefficients**.

The line over the conditional means



The line summarizes the conditional means

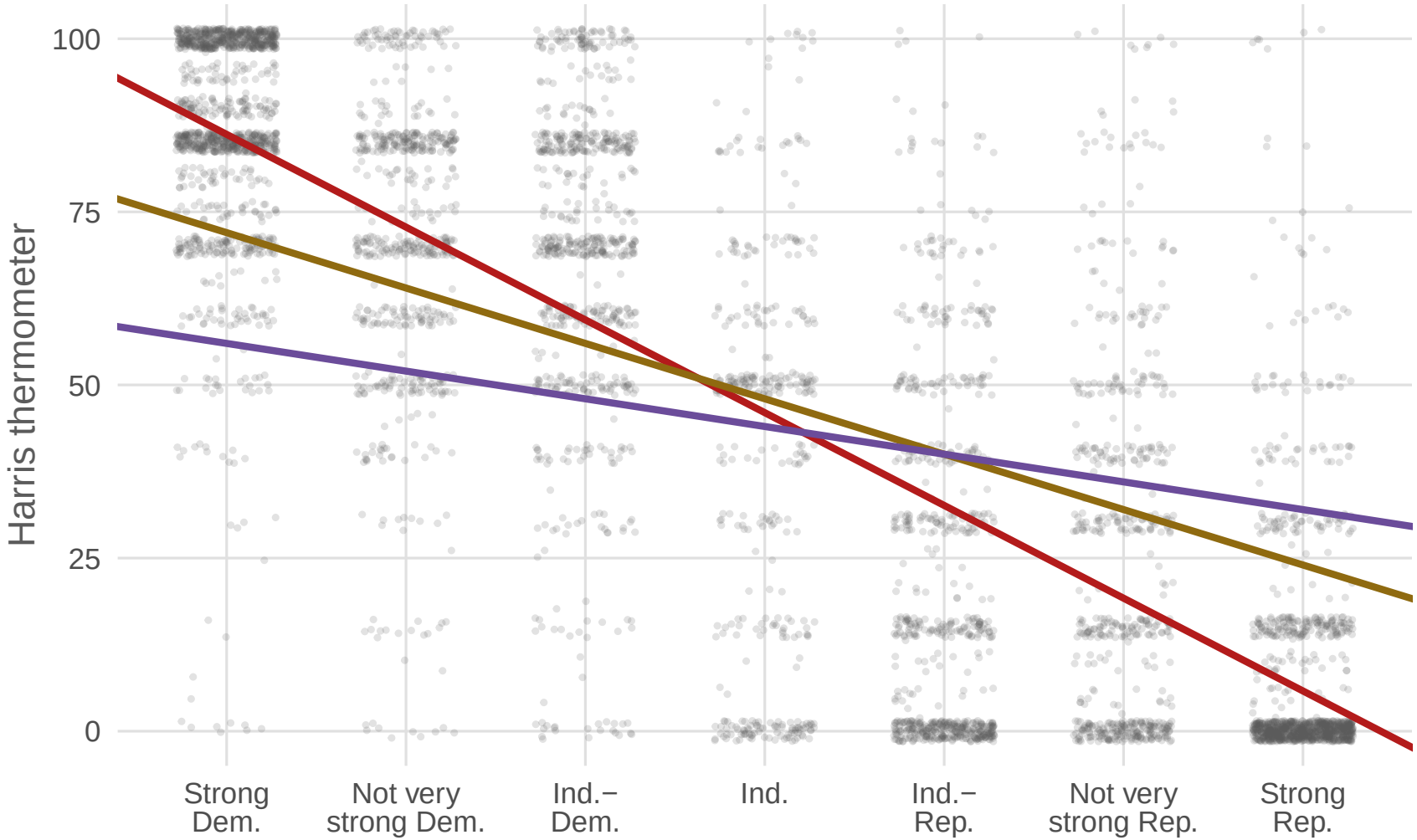


Gailmard on the line

"A linear regression smooths out the jagged edges and peaks of the scatterplot and the conditional mean function. Neither the scatterplot as a whole nor a linear regression simplifying its patterns should be understood as the one true depiction of a relationship; these descriptive devices are more useful in conjunction with each other than as substitutes."

Gailmard, *Statistical Modeling and Inference for Social Science*, p. 44

Which line?


Harris thermometer
100
75
50
25
0
Strong Dem.
Not very strong Dem.
Ind.–Dem.
Ind.
Ind.–Rep.
Not very strong Rep.
Strong Rep.

Roadmap

Relating two variables

Where we are

The difference in means is a regression

From conditional means to a line

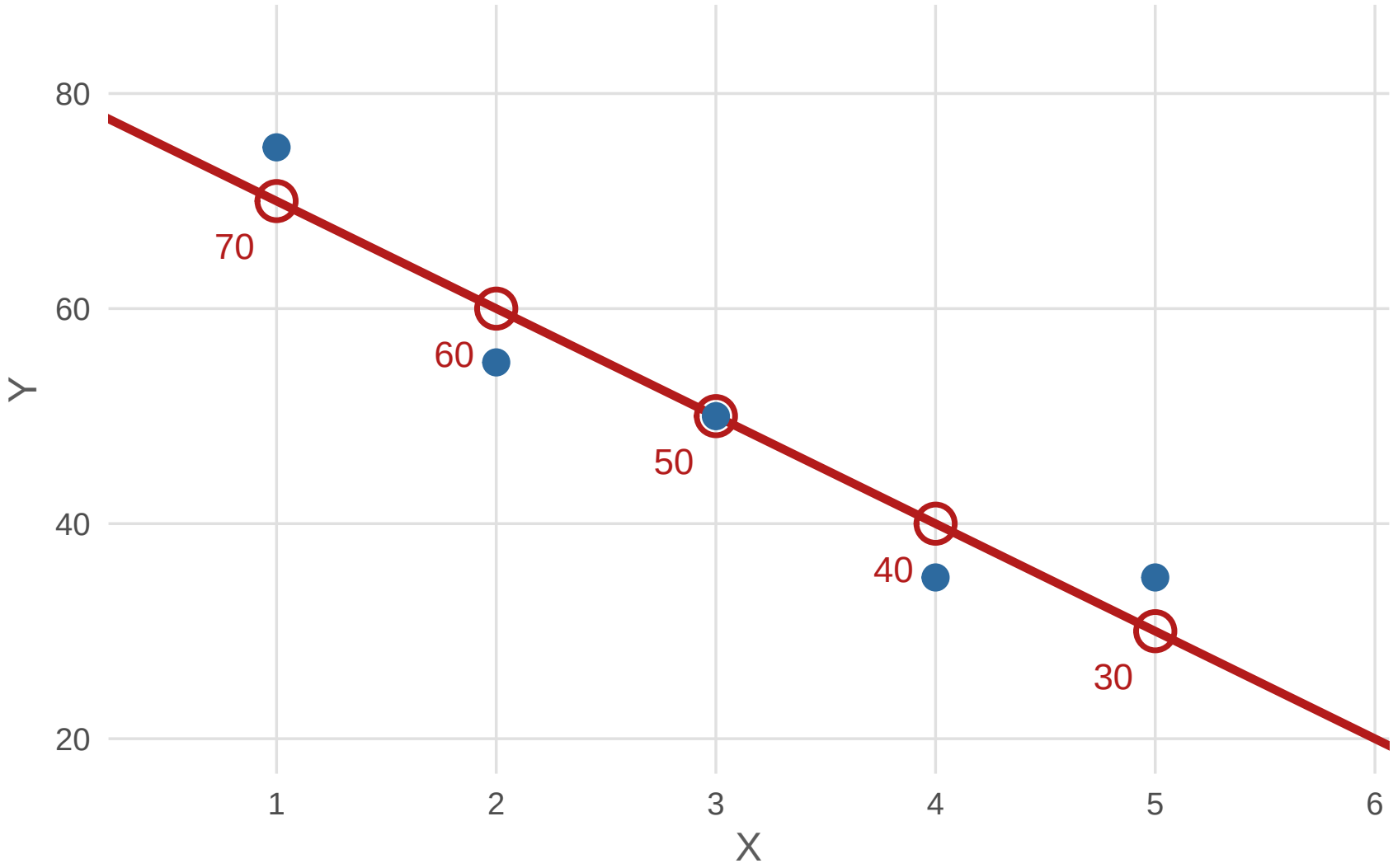
Least squares

Interpreting the line

How well does the line fit?

Wrapping up

Fitted values



Fitted values and residuals

FITTED VALUE AND RESIDUAL

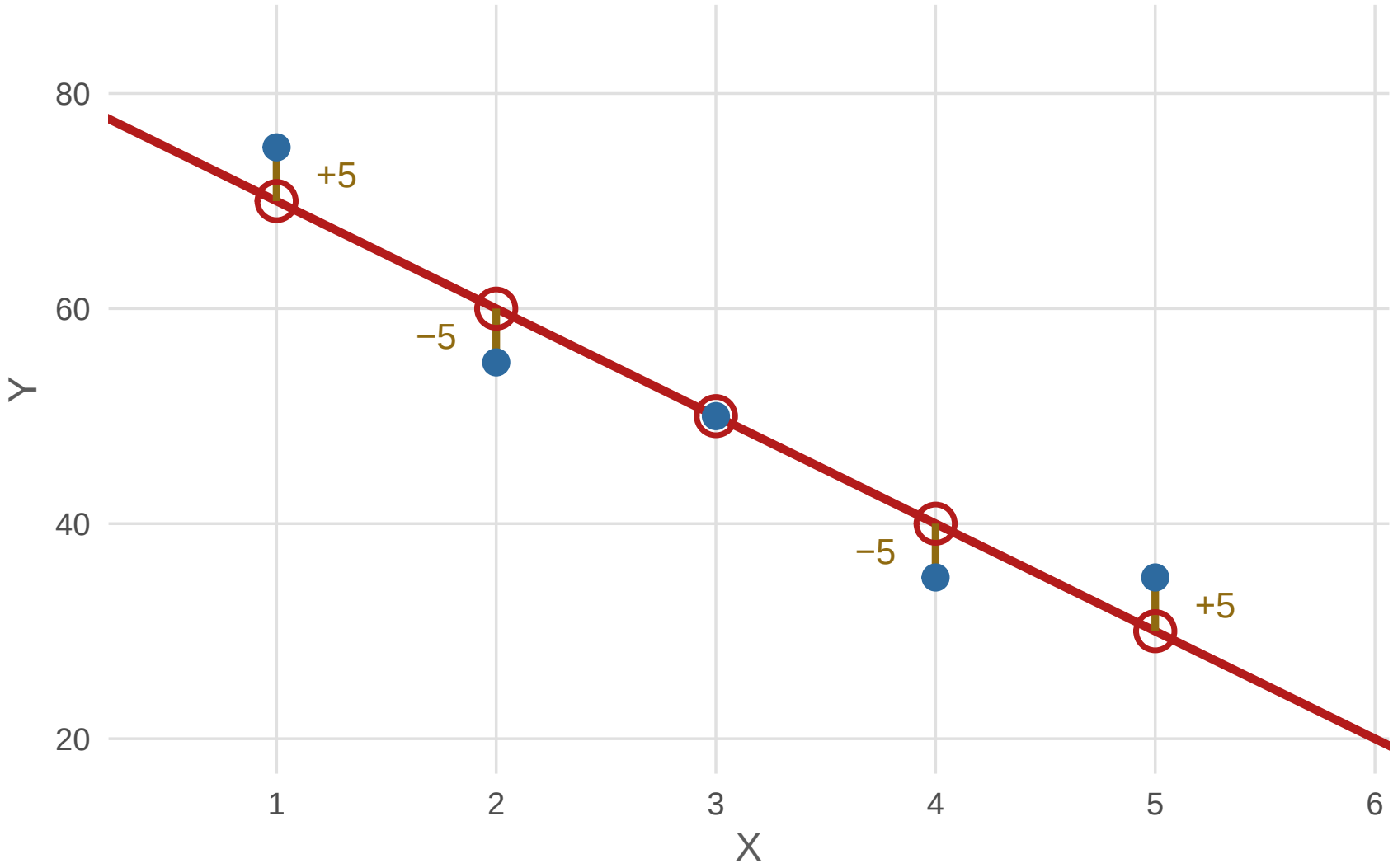
For a line with intercept b_0 and slope b_1, the **fitted value** for unit i is

$$\hat{Y}_i = b_0 + b_1 X_i,$$

and the **residual** is the "miss,"

$$\hat{\varepsilon}_i = Y_i - \hat{Y}_i.$$

Residuals



The “best” line minimizes those distances

- We can’t just minimize $\sum_i \hat{\varepsilon}_i$ because positive and negative residuals cancel out.
- So we *square* the residuals.
- When X takes at least two values, the squared criterion has exactly one best line.
- That line is our regression line.

From a constant to a line

BEST CONSTANT

$$\arg\min_{c} \sum_{i} (Y_i - c)^2$$

Answer: $\bar{Y}$

BEST LINE

$$\arg\min_{b_0, b_1} \sum_{i} (Y_i - b_0 - b_1 X_i)^2$$

$\hat{Y}_i = b_0 + b_1 X_i$, so

$$Y_i - b_0 - b_1 X_i = Y_i - \hat{Y}_i = \hat{\varepsilon}_i,$$

the residual.

Answer: $\hat{\beta}_0$ and $\hat{\beta}_1$, the **ordinary least squares** (OLS) coefficients

The OLS coefficients

The **sum of squared residuals** of a candidate line:

$$\text{SSR}(b_0, b_1) = \sum_{i=1}^{n} (Y_i - b_0 - b_1 X_i)^2 .$$

The OLS intercept and slope are the pair that makes SSR as small as possible:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg\min_{b_0, b_1} \text{SSR}(b_0, b_1).$$

BIVARIATE OLS

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X, Y)}{s_X^2}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Every step of the derivation: `handout_ols_derivation.pdf`.

Slope and correlation

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X, Y)}{s_X^2}$$

The OLS slope.

$$= \frac{\widehat{\text{Cov}}(X, Y)}{s_X \cdot s_X} \cdot \frac{s_Y}{s_Y}$$

Write $s_X^2 = s_X \cdot s_X$ and multiply by $s_Y / s_Y = 1$.

$$= \frac{\widehat{\text{Cov}}(X, Y)}{s_X \, s_Y} \cdot \frac{s_Y}{s_X}$$

Swap the two denominators.

$$= \hat{r} \frac{s_Y}{s_X}$$

The definition of $\hat{r}$.

In the bivariate case, the slope is the correlation, rescaled from standard deviations into the units of Y per unit of X.

A toy example

i	X_i	Y_i	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X})(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$
1	1	75	−2	25	−50	4
2	2	55	−1	5	−5	1
3	3	50	0	0	0	0
4	4	35	1	−15	−15	1
5	5	35	2	−15	−30	4
Mean	3	50				
Sum			0	0	**−100**	**10**

$$\widehat{\text{Cov}}(X, Y) = \frac{-100}{5-1} = -25 \qquad s_X^2 = \frac{10}{5-1} = 2.5$$

The toy example, solved

$$\widehat{\text{Cov}}(X, Y) = \frac{1}{n-1}\sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = \frac{-100}{4} = -25$$

$$s_X^2 = \frac{1}{n-1}\sum_i (X_i - \bar{X})^2 = \frac{10}{4} = 2.5$$

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X, Y)}{s_X^2} = \frac{-25}{2.5} = -10$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 50 - (-10)(3) = 80$$

$$\hat{Y}_i = 80 - 10\, X_i$$

The covariance and the variance both divide by $n - 1$, so $n - 1$ cancels from the ratio:

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{-100}{10} = -10$$

The toy example in R

```
toy <- data.frame(X = 1:5, Y = c(75, 55, 50, 35, 35))
coef(lm(Y ~ X, data = toy))
```

```
(Intercept)           X
         80         -10
```

Party ID: the formula and `lm()`

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X, Y)}{s_X^2}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

THE FORMULA

```
d  <- na.omit(
  anes[c("pid", "harris")])
b1 <- cov(d$pid, d$harris) /
      var(d$pid)
b0 <- mean(d$harris) -
      b1 * mean(d$pid)
c(b0, b1)
```

```
[1]  99.591 -13.398
```

`lm()`

```
fit <- lm(harris ~ pid,
          data = anes)
coef(fit)
```

```
(Intercept)         pid
     99.591     -13.398
```

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The slope

INTERPRETING $\hat{\beta}_1$

$$\widehat{\texttt{harris}} = 99.59 - 13.40\,\texttt{pid}$$

On average, a one-unit increase in party ID is associated with a 13.4-point decrease in the Harris thermometer rating.

In general: on average, a one-unit increase in X is associated with a $\hat{\beta}_1$-unit change in Y.

Interpreting $\hat{\beta}_1$: check yourself

- What is a one-unit increase in party ID?

  One step on the 7-point scale, toward Republican: Strong Democrat (1) to Not very strong Democrat (2).

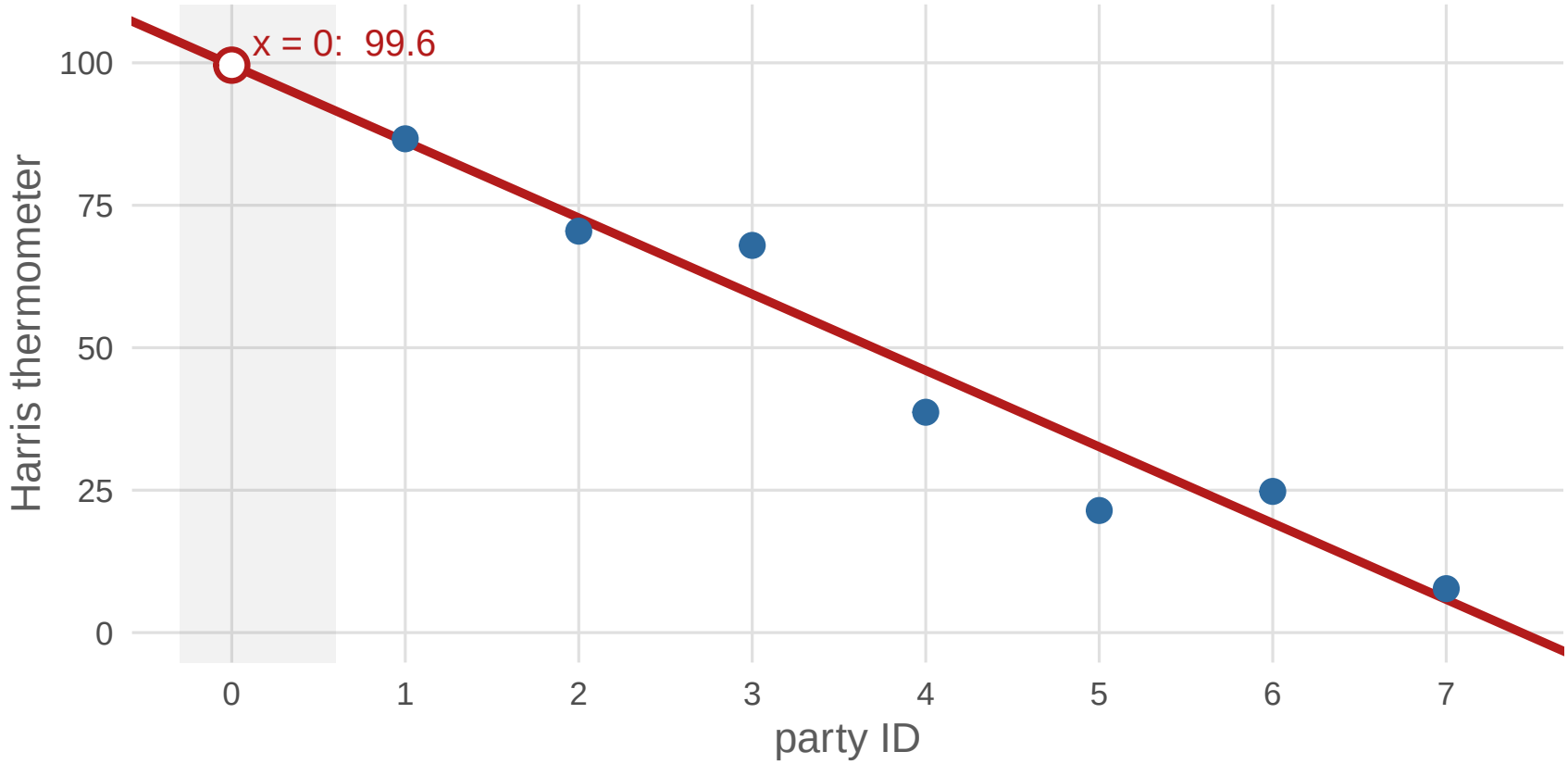
- A Strong Democrat (1) and a Strong Republican (7): how far apart are their predicted ratings?

  Six steps: $6 \times 13.4 = 80.4$ points.

- Does moving a respondent one step toward Republican lower their rating of Harris by 13.4 points?

  Careful! We say "is associated with", not "causes".

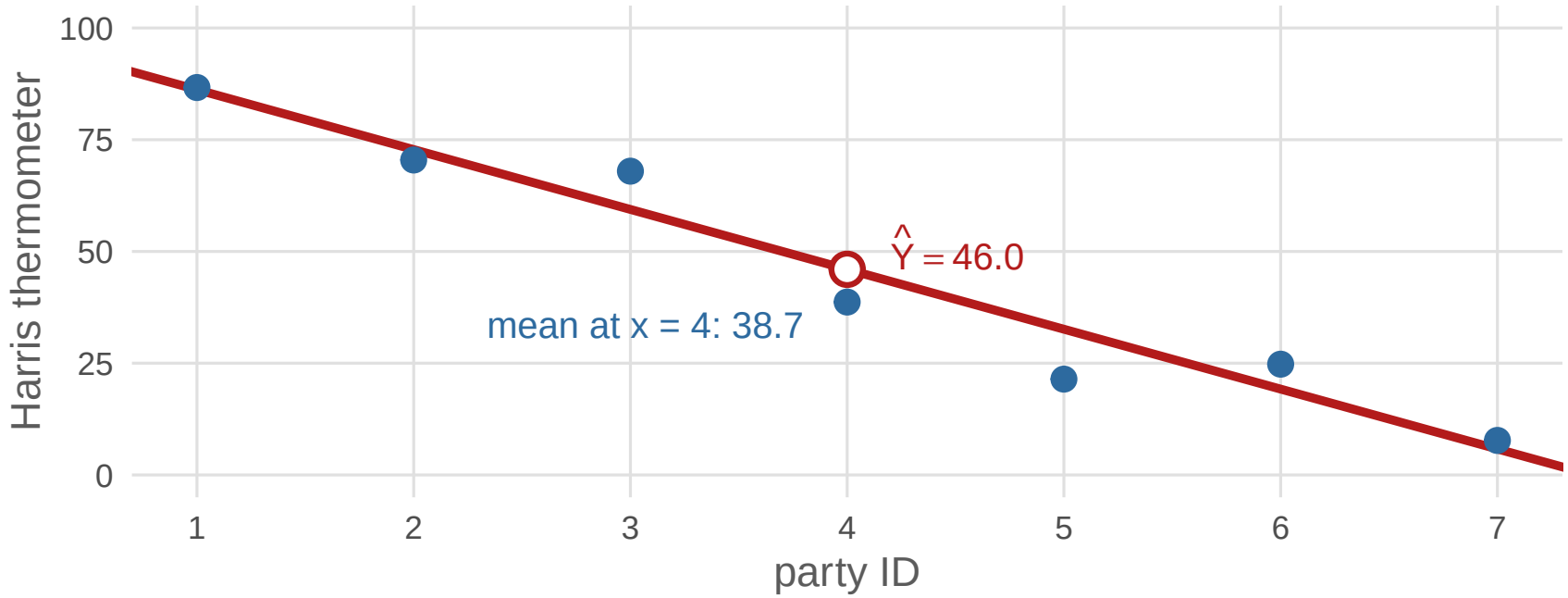
The intercept



$\hat{\beta}_0$ = 99.6 is the predicted rating at party ID = 0. What does this mean?

A prediction from the line

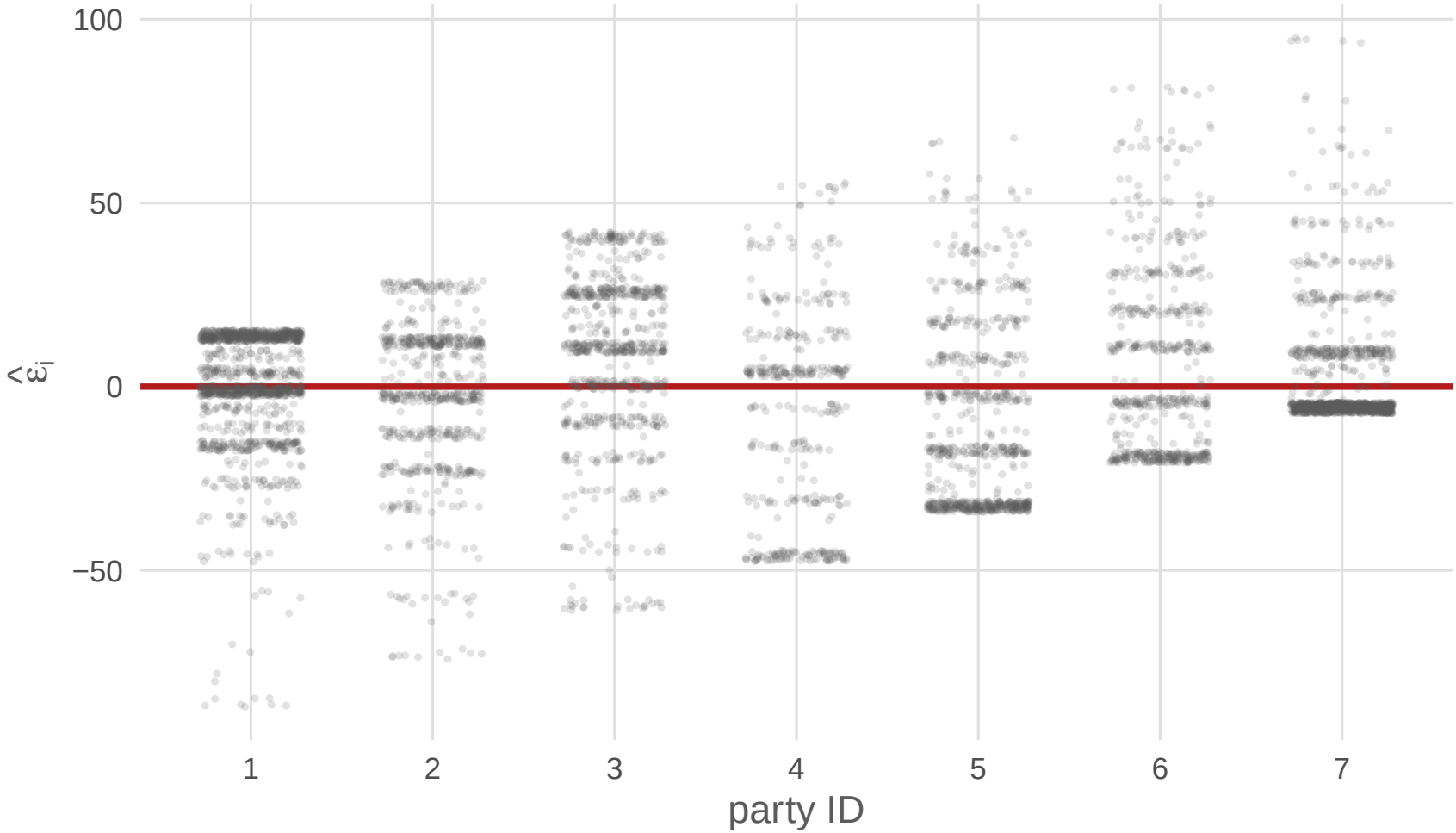
What is the predicted Harris rating for an Independent (`pid = 4`)?



$$\widehat{\texttt{harris}} = 99.59 - 13.40\,\texttt{pid}$$

$$\hat{Y} = 99.59 - 13.40 \times 4 = 46.0$$

The residuals



OLS residuals average to zero

OLS RESIDUALS

$$\sum_i \hat{\varepsilon}_i = 0 \qquad \text{and} \qquad \sum_i X_i \hat{\varepsilon}_i = 0.$$

The residuals average to zero, and they have no linear relationship with X. The line has "used up" all the linear information in X.

The line passes through $(\bar{X}, \bar{Y})$

$$\hat{\beta}_0 + \hat{\beta}_1 \bar{X}$$

The line at $X = \bar{X}$.

$$= (\bar{Y} - \hat{\beta}_1 \bar{X}) + \hat{\beta}_1 \bar{X}$$

Substitute $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$.

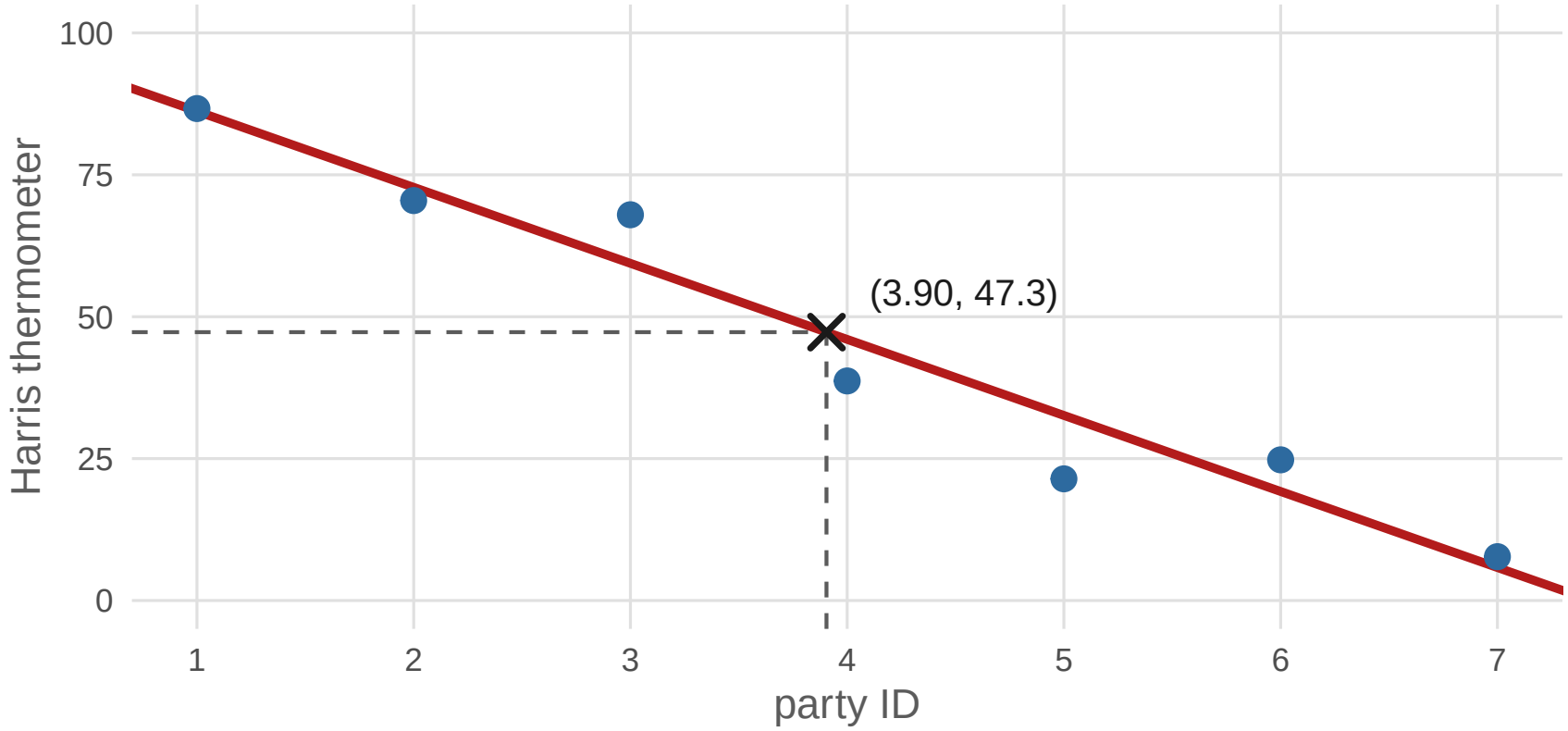
$$= \bar{Y}$$

The point $(\bar{X}, \bar{Y})$ is on the line.

The means in our example

In our example, $\bar{X} = 3.90$ and $\bar{Y} = 47.3$. The line at $\bar{X}$:

$$99.59 - 13.40 \times 3.90 = 47.3.$$



Units

AGE IN YEARS

```
coef(lm(harris ~ age,
        data = anes))
```

```
(Intercept)         age
     41.685       0.111
```

AGE IN DECADES

```
coef(lm(harris ~ I(age / 10),
        data = anes))
```

```
(Intercept)   I(age/10)
     41.685       1.111
```

$\hat{\beta}_1$ is in units of Y per unit of X. Dividing X by 10 multiplies $\hat{\beta}_1$ by 10; $\hat{\beta}_0$ stays the same.

The fitted values do not change: a 50-year-old gets the same $\hat{Y}$ from either line.

Describing, not explaining

DESCRIPTIVE

"is associated with"
"differ by"

On average, respondents one step more Republican rate Harris 13.4 points lower.

CAUSAL

"the effect of"
"increases" / "reduces"
"leads to"

Becoming one step more Republican lowers a person's Harris rating by 13.4 points.

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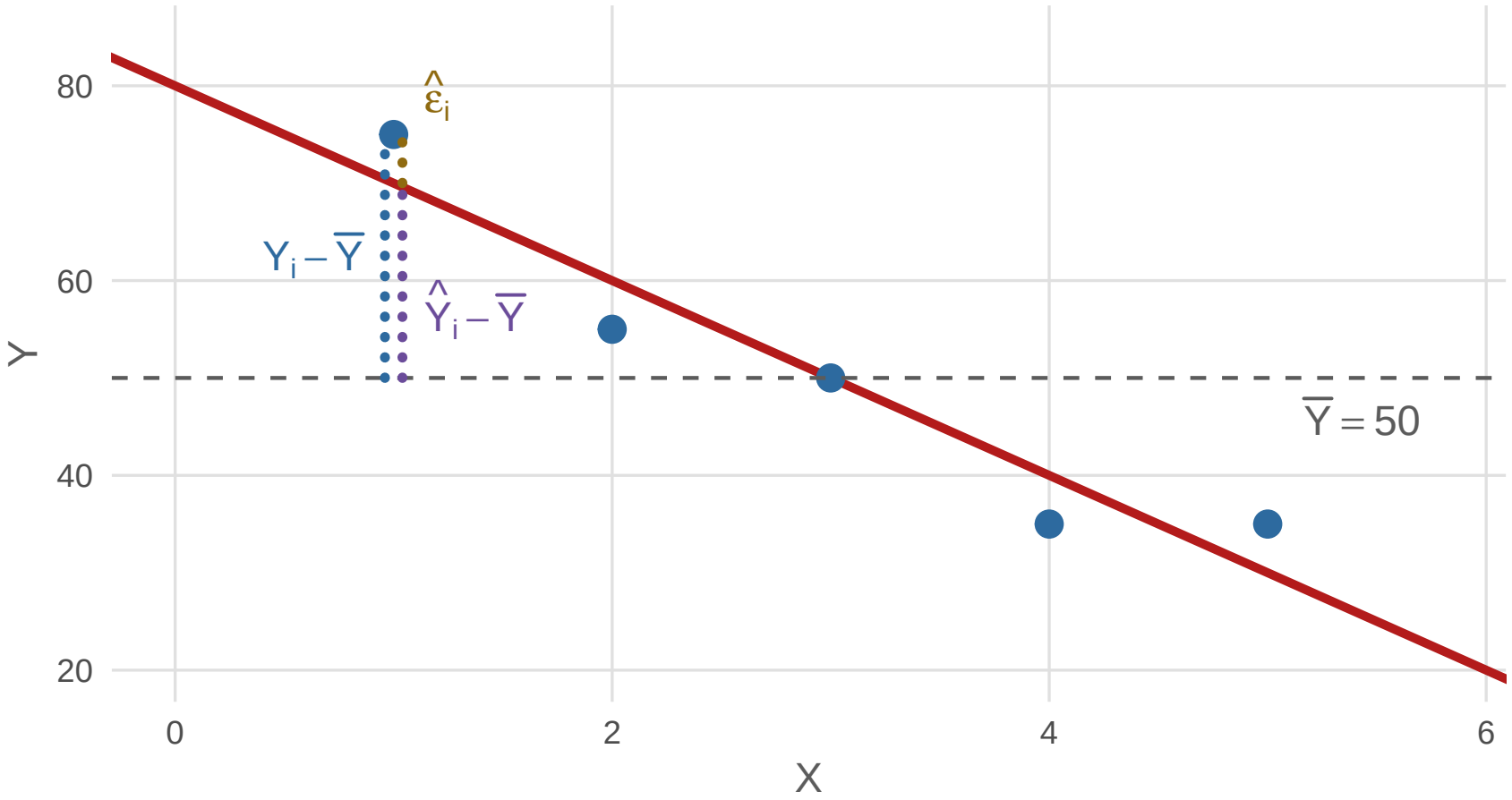
Least squares

Interpreting the line

How well does the line fit?

Wrapping up

One respondent's deviation



$$Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + \hat{\varepsilon}_i$$

Three sums of squares

SUMS OF SQUARES

The **total sum of squares**, the **explained sum of squares** and the **sum of squared residuals**:

$$\text{TSS} = \sum_i (Y_i - \bar{Y})^2, \quad \text{ESS} = \sum_i (\hat{Y}_i - \bar{Y})^2, \quad \text{SSR} = \sum_i \hat{\varepsilon}_i^2.$$

$$\text{TSS} = \text{ESS} + \text{SSR}$$

COEFFICIENT OF DETERMINATION

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{SSR}}{\text{TSS}}$$

"$R^2 \times 100\%$ of the variation in Y is accounted for by X."
$0 \leq R^2 \leq 1$.

R^2 and the correlation

BIVARIATE REGRESSION

$$R^2 = \hat{r}^2$$

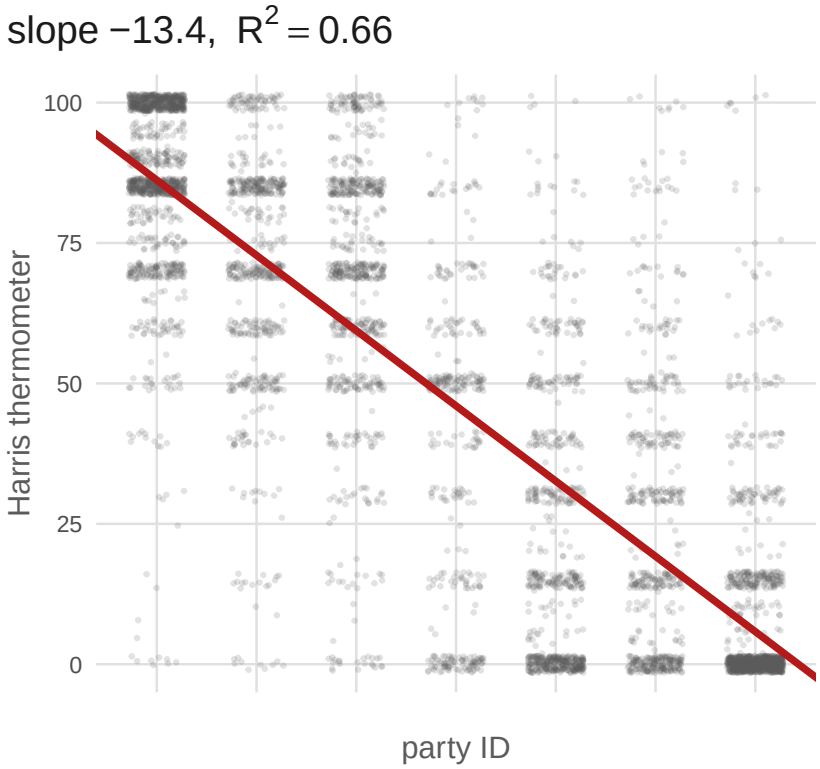
In our running example, the Harris feeling thermometer and `pid`:

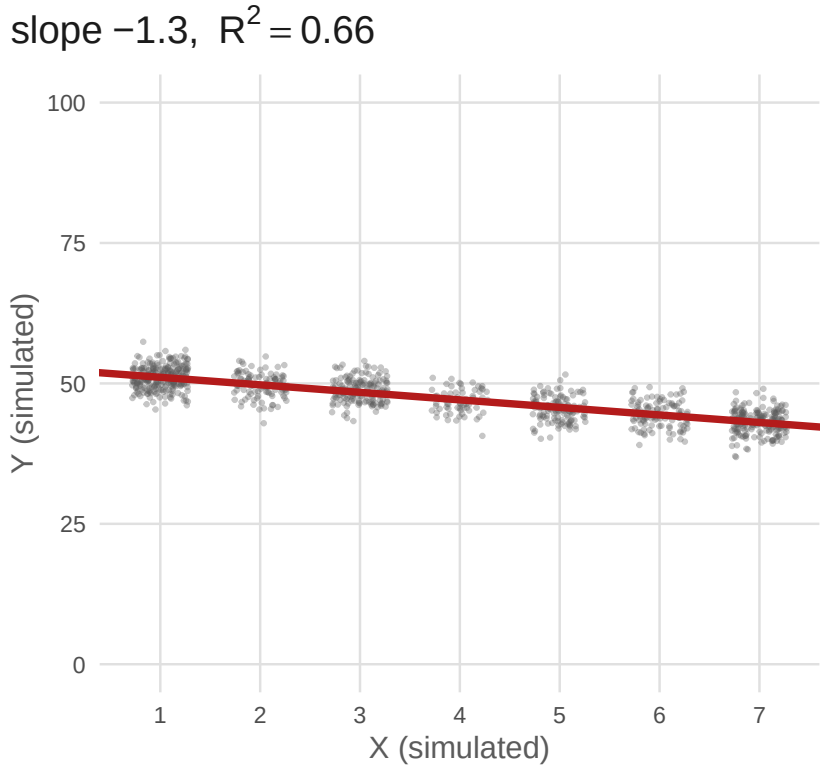
$$\hat{r} = -0.81, \qquad R^2 = (-0.81)^2 = 0.66.$$

66% of the variation in the Harris feeling thermometer is accounted for by `pid`.

Squaring drops the sign: $\hat{r} = 0.81$ would give the same R^2.

Slope size is not fit





What R^2 does not tell you

- Whether a straight line is the right shape for the relationship.
- Whether the slope is substantively large.
- Anything about cause and effect.
- Whether a value is “large” or “small”: there is no universal standard.

Interpreting `summary()`

```
summary(lm(harris ~ pid, data = anes))
```

```
Call:
lm(formula = harris ~ pid, data = anes)

Residuals:
    Min      1Q  Median      3Q     Max
-86.193  -9.398  -1.193  13.807  94.193

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  99.5907     0.5845   170.4   <2e-16 ***
pid         -13.3977     0.1293  -103.6   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 21.79 on 5465 degrees of freedom
  (54 observations deleted due to missingness)
Multiple R-squared:  0.6627,  Adjusted R-squared:  0.6627
F-statistic: 1.074e+04 on 1 and 5465 DF,  p-value: < 2.2e-16
```

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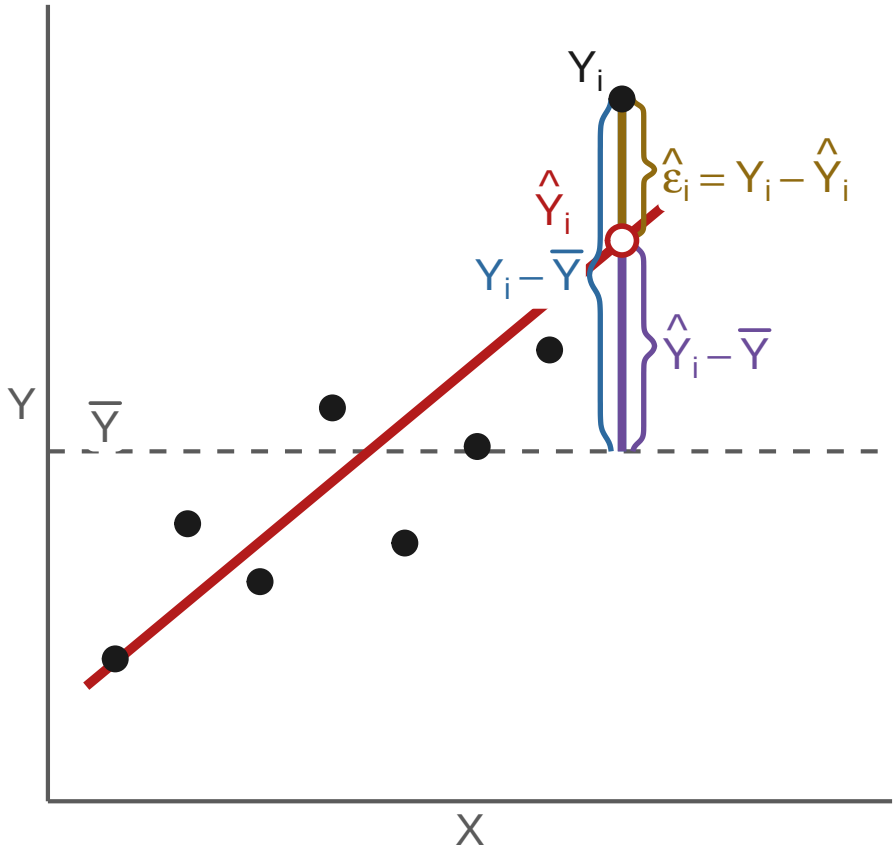
Least squares

Interpreting the line

How well does the line fit?

Wrapping up

Splitting one deviation



$\bar{Y}$: the mean, the best constant prediction

$Y_i - \bar{Y}$: the total deviation, from the mean

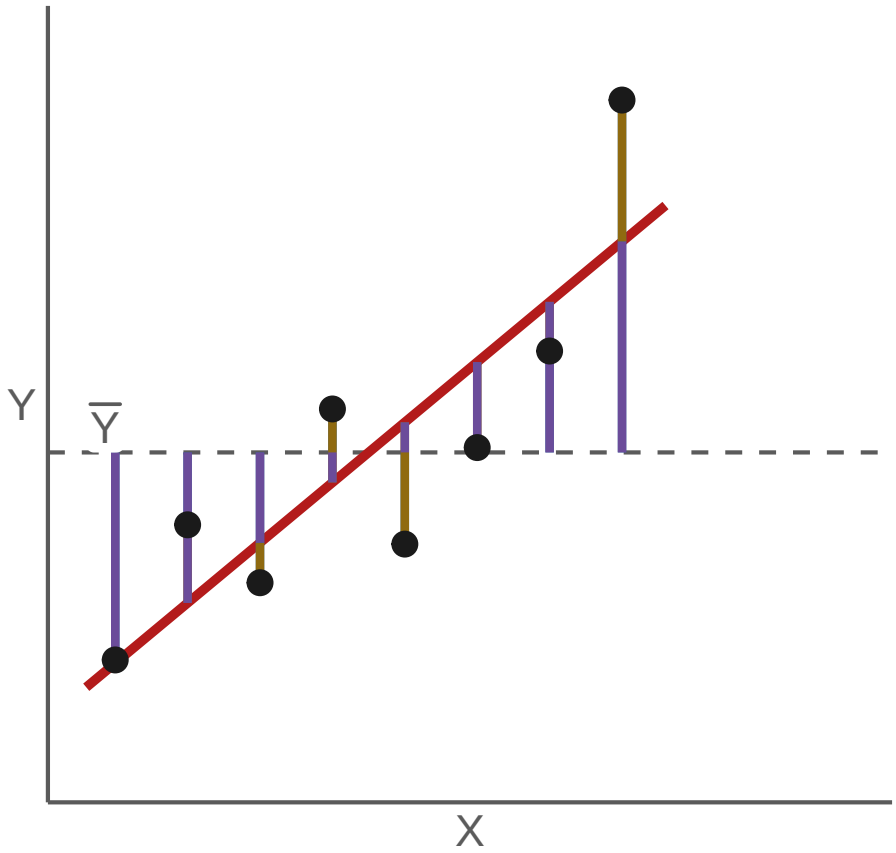
$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$: the fitted value, on the line

$\hat{\varepsilon}_i = Y_i - \hat{Y}_i$: the unexplained deviation, or residual

$\hat{Y}_i - \bar{Y}$: the explained deviation

$$Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + \hat{\varepsilon}_i$$

Adding up the squares



Total sum of squares:
$\text{TSS} = \sum_i (Y_i - \bar{Y})^2$
The numerator of the variance:
$s_Y^2 = \text{TSS}/(n-1)$.

Sum of squared residuals:
$\text{SSR} = \sum_i (Y_i - \hat{Y}_i)^2$
$= \sum_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$
OLS picks the line with the smallest SSR: the least total squared vertical distance from the points.

Explained (or regression) sum of squares:
$\text{ESS} = \sum_i (\hat{Y}_i - \bar{Y})^2$

$\text{TSS} = \text{ESS} + \text{SSR}$

What the line accounts for

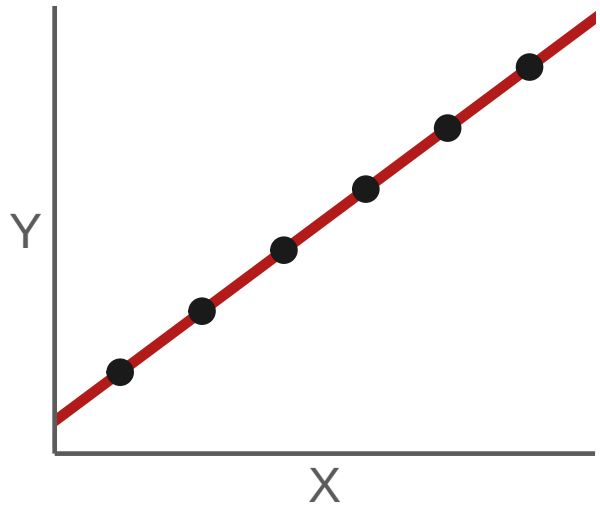
$$R^2 = \frac{\text{TSS} - \text{SSR}}{\text{TSS}} = 1 - \frac{\text{SSR}}{\text{TSS}}$$

$100 \times R^2$: the percent of the variation in Y that X accounts for

If SSR $= 0$, every point sits on the line. Then $R^2 = 1$.

Bivariate regression: $R^2 = \hat{r}^2$

$\hat{\beta}_1$: the average difference in Y between units one unit apart on X



Three best predictions

Prediction	Criterion	Solution
One number	$\arg\min_{c} \sum_i (Y_i - c)^2$	$\bar{Y}$
A line, $X \in \{0, 1\}$	$\arg\min_{b_0, b_1} \sum_i (Y_i - b_0 - b_1 X_i)^2$	$\hat{\beta}_0 = \bar{Y}_{X=0}$ $\hat{\beta}_1 = \bar{Y}_{X=1} - \bar{Y}_{X=0}$
A line, any X	$\arg\min_{b_0, b_1} \sum_i (Y_i - b_0 - b_1 X_i)^2$	$\hat{\beta}_1 = \widehat{\text{Cov}}(X, Y)/s_X^2$ $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$

Further reading

- Gailmard, *Statistical Modeling and Inference for Social Science*, section 2.3.5.
- Aronow and Miller, *Foundations of Agnostic Statistics*, section 4.1.

These slides draw on teaching materials by Talbot Andrews, Daniel de Kadt, Tom Pepinsky and Teppei Yamamoto.